Sequences and Basic Series Convergence Tests

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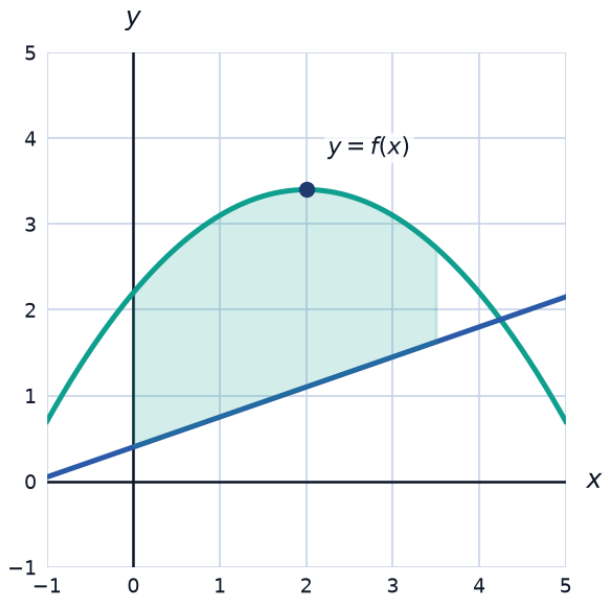
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Limits of Sequences & Monotonicity**

Find the interval of convergence of $\sum_{n=0}^{\infty} x^n$.

2 **[Foundational] • Target: Geometric Series & n-th Term Divergence Test**

The graph of the function $f(x)$ is shown below. What is the value of the definite integral $\int_{-2}^{2} f(x)\,dx$?



3 **[Foundational] • Target: Integral Test & p-Series**

Use a two-term Maclaurin polynomial for e^x to approximate $e^{0.1}$.

4 **[Foundational] • Target: Direct Comparison & Limit Comparison Tests**

Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{2^n}$.

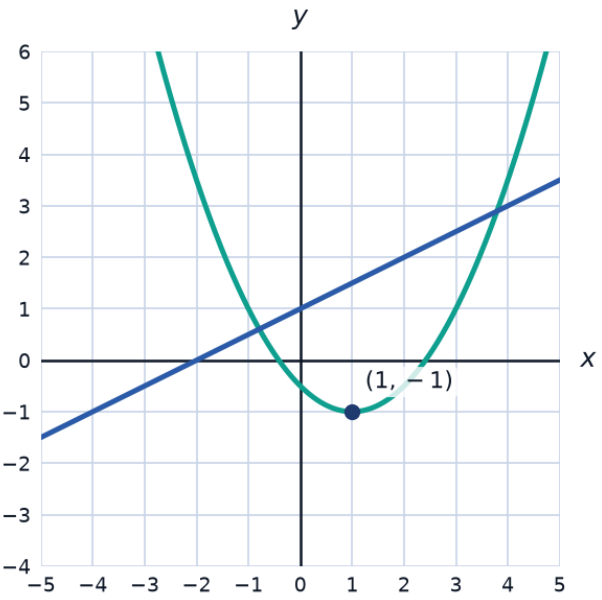
5 **[Foundational] • Target: Limits of Sequences & Monotonicity**

Consider the function represented by the graph of $f(x)$ below. What is the value of the definite integral $\int_{0}^{4} f(x)\,dx$?

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6 [Foundational] • Target: Geometric Series & n-th Term Divergence Test

Consider the infinite series $\sum_{n=1}^{\infty} \frac{1}{n^p}$. For what value(s) of p does this series converge?



7 [Foundational] • Target: Integral Test & p-Series

Find the arc length of $x = t,\ y = t$ for $t \in [0, 1]$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Direct Comparison & Limit Comparison Tests

Evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

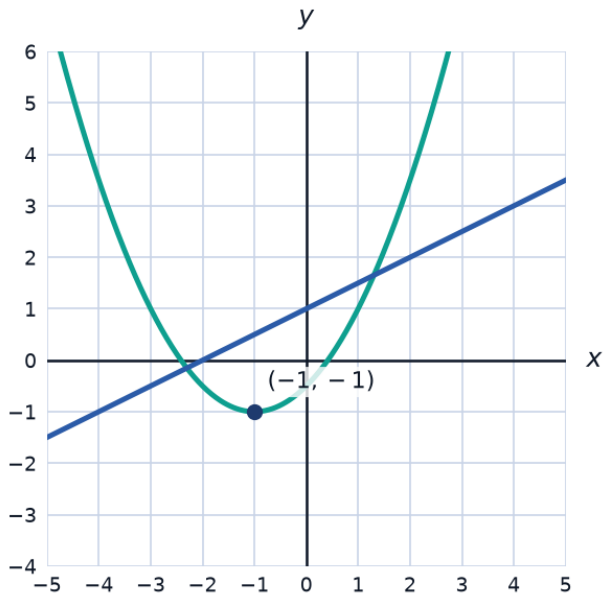
9 [Intermediate] • Target: Limits of Sequences & Monotonicity

Does $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converge?

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10 [Intermediate] • Target: Geometric Series & n-th Term Divergence Test

Find the speed of a particle with $x = t^2,\ y = t^2$ at $t = 1$.



11 [Intermediate] • Target: Integral Test & p-Series

Evaluate $\int \frac{1}{x^2 - 1}\,dx$.

12 [Intermediate] • Target: Direct Comparison & Limit Comparison Tests

Does $\sum_{n=1}^{\infty} \frac{1}{n}$ converge?

13 [Intermediate] • Target: Limits of Sequences & Monotonicity

Does $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converge or diverge?

14 [Intermediate] • Target: Geometric Series & n-th Term Divergence Test

Does $\sum_{n=1}^{\infty} \frac{n}{n+1}$ converge?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

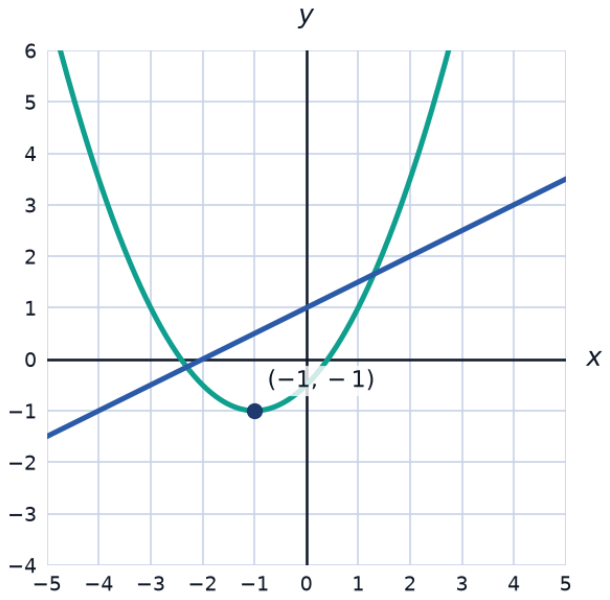
15 [Challenging] • Target: Integral Test & p-Series • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

For $x = t^2,\ y = t^3$, find $\frac{dy}{dx}$.

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16 [Challenging] • Target: Direct Comparison & Limit Comparison Tests • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Evaluate $\int_0^{\infty} e^{-x}\,dx$.



17 [Challenging] • Target: Limits of Sequences & Monotonicity • Cross-Lesson: 3.01 Parametric Equations

Evaluate $\int x\,e^{x^2}\,dx$.

18 [Challenging] • Target: Geometric Series & n-th Term Divergence Test • Cross-Lesson: 3.02 Vector-Valued Functions

Evaluate $\sum_{n=0}^{\infty} (\frac{1}{2})^n$.

19 [Challenging] • Target: Integral Test & p-Series • Cross-Lesson: 4.01 Polar Curves and Derivatives

Evaluate $\int x\,e^{x}\,dx$.

20 [Challenging] • Target: Direct Comparison & Limit Comparison Tests • Cross-Lesson: 4.02 Area in Polar Coordinates

For $x = e^t,\ y = e^{2t}$, find $\frac{dy}{dx}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Limits of Sequences & Monotonicity

Determine the limit of the sequence $a_n = \frac{4n^2 - 3n}{2n^2 + 5}$.

a. 2

b. 4

c. ∞ (diverges)

d. 0

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22 [Foundational] • Target: Limits of Sequences & Monotonicity

Which statement is guaranteed by the Monotone Convergence Theorem for Sequences?

a. Every bounded monotonic sequence converges

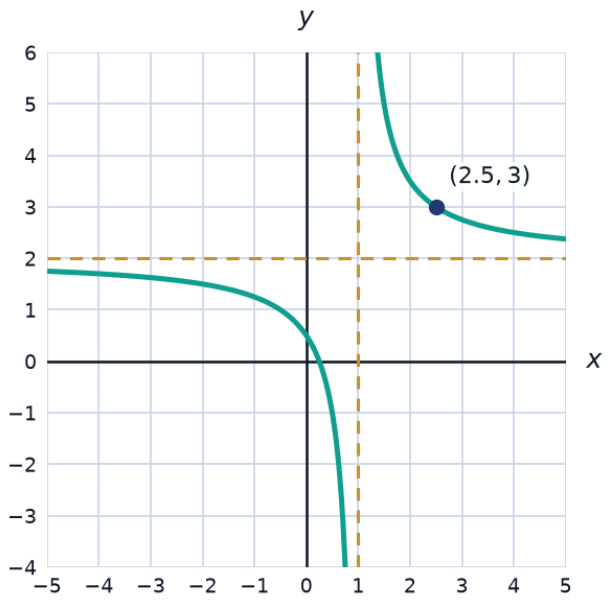
b. Every alternating sequence converges

c. Every bounded sequence is monotonic

d. Every convergent sequence is positive

23 [Foundational] • Target: Limits of Sequences & Monotonicity

What is $\lim_{n\to\infty}(1+\frac{2}{n})^n$?



a. e^2

b. e

c. 1

d. ∞

24 [Foundational] • Target: Geometric Series & n-th Term Divergence Test

Evaluate the sum of the series $\sum_{n=1}^{\infty} 3(\frac{2}{5})^n$.

a. 2

b. $\frac{6}{5}$

c. $\frac{15}{2}$

d. $\frac{5}{2}$

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25 [Foundational] • Target: Geometric Series & n-th Term Divergence Test

If $\lim_{n\to\infty} a_n = \frac{1}{2}$, what does the n-th Term Test conclude about $\sum_{n=1}^{\infty} a_n$?

a. The series diverges

b. The series converges to $\frac{1}{2}$

c. The test is inconclusive

d. The series is geometric

26 [Foundational] • Target: Geometric Series & n-th Term Divergence Test

If $\lim_{n\to\infty} a_n = 0$, what can be concluded about $\sum a_n$?

a. The test is inconclusive; the series may converge or diverge

b. The series converges

c. The series diverges

d. The sum is zero

27 [Foundational] • Target: Integral Test & p-Series

For what values of p does the p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converge?

a. $p > 1$

b. $p \geq 1$

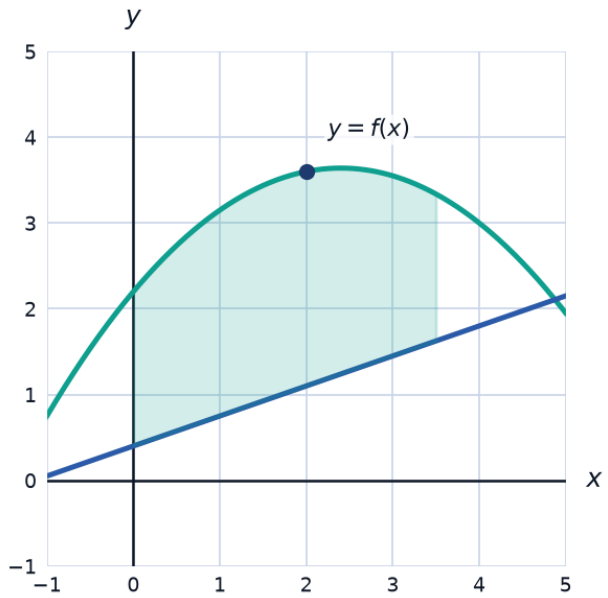
c. $p < 1$

d. $p > 0$

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28 [Foundational] • Target: Integral Test & p-Series

Which series converges by the Integral Test?



a. $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$

b. $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$

c. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

d. $\sum_{n=1}^{\infty} \frac{1}{2n+1}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Integral Test & p-Series

What three conditions must $f(x)$ satisfy on $[1, \infty)$ to apply the Integral Test to $\sum a_n$?

a. $f(x)$ must be continuous, positive, and decreasing

b. $f(x)$ must be differentiable, alternating, and bounded

c. $f(x)$ must be positive and increasing

d. $f(x)$ must be a rational function

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30 [Intermediate] • Target: Direct Comparison & Limit Comparison Tests

Determine the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^3+5n}$.

a. Converges by Direct Comparison with $\sum \frac{1}{n^3}$

b. Diverges by Comparison with $\sum \frac{1}{n}$

c. Diverges by n-th term test

d. Inconclusive

31 [Intermediate] • Target: Direct Comparison & Limit Comparison Tests

Which series should be used for the Limit Comparison Test with $\sum_{n=1}^{\infty} \frac{3n+2}{2n^3-1}$?

a. $\sum \frac{1}{n^2}$ (convergent)

b. $\sum \frac{1}{n}$ (divergent)

c. $\sum \frac{1}{n^3}$ (convergent)

d. $\sum n$ (divergent)

32 [Intermediate] • Target: Direct Comparison & Limit Comparison Tests

In the Limit Comparison Test, if $\lim_{n\to\infty} \frac{a_n}{b_n} = L$ where $0 < L < \infty$, what does this mean?

a. Both $\sum a_n$ and $\sum b_n$ either both converge or both diverge

b. $\sum a_n$ converges to L

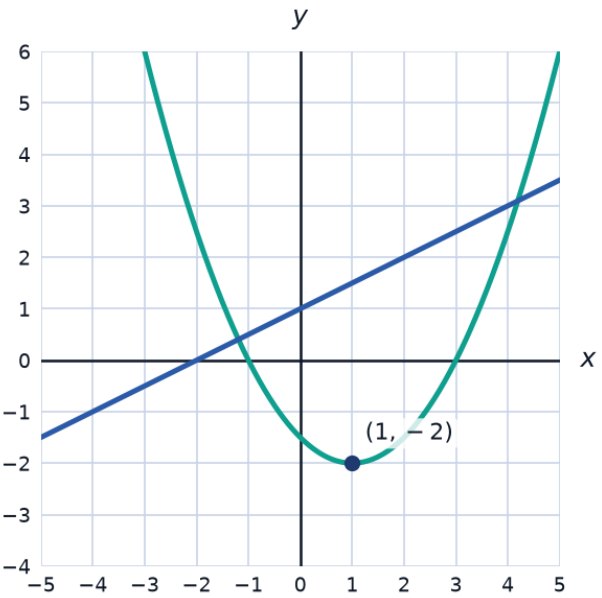
c. $\sum a_n$ converges while $\sum b_n$ diverges

d. The test is inconclusive

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33 [Intermediate] • Target: Limits of Sequences & Monotonicity

Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+6}$ converges.



a. Diverges by the nth – term test

b. Converges (compare to $\sum 1/n^2$)

c. Diverges (compare to $\sum 1/n$)

d. Inconclusive

34 [Intermediate] • Target: Geometric Series & n-th Term Divergence Test

Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+5}$ converges.

a. Inconclusive

b. Diverges by the nth – term test

c. Diverges (compare to $\sum 1/n$)

d. Converges (compare to $\sum 1/n^2$)

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35 **[Intermediate]** • **Target: Integral Test & p-Series**

Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+3}$ converges.

a. Diverges (compare to $\sum 1/n$)

b. Inconclusive

c. Diverges by the nth – term test

d. Converges (compare to $\sum 1/n^2$)

36 **[Intermediate]** • **Target: Direct Comparison & Limit Comparison Tests**

Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+2}$ converges.

a. Converges (compare to $\sum 1/n^2$)

b. Diverges by the nth – term test

c. Diverges (compare to $\sum 1/n$)

d. Inconclusive

37 **[Intermediate]** • **Target: Limits of Sequences & Monotonicity**

Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+4}$ converges.

a. Diverges (compare to $\sum 1/n$)

b. Diverges by the nth – term test

c. Converges (compare to $\sum 1/n^2$)

d. Inconclusive

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Geometric Series & n-th Term Divergence Test • Cross-Lesson: 3.02 Vector-Valued Functions

Consider the infinite series $\sum_{n=1}^{\infty} a_n$. If the terms of the series are given by $a_n = \frac{3n+1}{2n+5}$, which of the following statements about the convergence of the series is true based on the limit of the terms?

a. The series converges because $\lim_{n\to\infty} a_n = 0$.

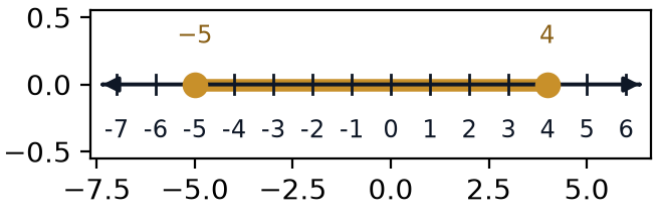
b. The series diverges because $\lim_{n\to\infty} a_n = \frac{3}{2}$.

c. The series converges because $\lim_{n\to\infty} a_n = \frac{3}{2}$.

d. The series diverges because $\lim_{n\to\infty} a_n$ does not exist.

39 [Challenging] • Target: Integral Test & p-Series • Cross-Lesson: 4.01 Polar Curves and Derivatives

Consider the infinite series $\sum_{n=1}^{\infty} \frac{n^p}{5^n}$. The series converges if and only if the parameter p satisfies which of the following conditions? Use the Ratio Test to determine the interval of convergence.



a. $p > 0$

b. $p < 1$

c. $p > -1$

d. For all real values of p

40 [Challenging] • Target: Direct Comparison & Limit Comparison Tests • Cross-Lesson: 4.02 Area in Polar Coordinates

Consider the infinite sequence defined by $a_n = \frac{3n^2 + 2n - 1}{n^2 + 5}$. To determine if this sequence converges, we examine its limit as $n \to \infty$. Which of the following statements is true about the limit of a_n?

a. The limit is 0.

b. The limit is 2.

c. The limit is 3.

d. The sequence diverges.

41 [Challenging] • Target: Limits of Sequences & Monotonicity • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Geometric series $\sum ar^n$ converges when:

a. $r > 1$

b. $|r| < 1$

c. $r < 0$

d. $r = 1$

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42 [Challenging] • Target: Geometric Series & n-th Term Divergence Test • Cross-Lesson: 6.01 Power Series and Interval of Convergence

For $\sum_{n=1}^{\infty} \frac{n^{13}}{2^n}$, apply the ratio test and determine convergence.

a. Converges ($L = 1/2 < 1$)

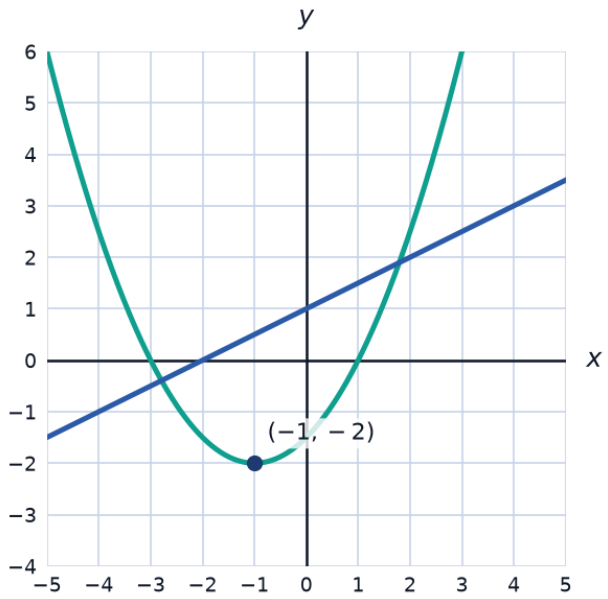
b. Inconclusive ($L = 1$)

c. Diverges by comparison

d. Diverges ($L = 2 > 1$)

43 [Challenging] • Target: Integral Test & p-Series • Cross-Lesson: 6.02 Taylor and Maclaurin Series

For $\sum_{n=1}^{\infty} \frac{n^{12}}{2^n}$, apply the ratio test and determine convergence.



a. Diverges by comparison

b. Diverges ($L = 2 > 1$)

c. Converges ($L = 1/2 < 1$)

d. Inconclusive ($L = 1$)

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44 [Challenging] • Target: Direct Comparison & Limit Comparison Tests • Cross-Lesson: 1.01 Integration by Parts

For $\sum_{n=1}^{\infty} \frac{n^{10}}{2^n}$, apply the ratio test and determine convergence.

a. Diverges by comparison

b. Inconclusive ($L = 1$)

c. Diverges ($L = 2 > 1$)

d. Converges ($L = 1/2 < 1$)

45 [Challenging] • Target: Limits of Sequences & Monotonicity • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

For $\sum_{n=1}^{\infty} \frac{n^{11}}{2^n}$, apply the ratio test and determine convergence.

a. Diverges ($L = 2 > 1$)

b. Diverges by comparison

c. Inconclusive ($L = 1$)

d. Converges ($L = 1/2 < 1$)

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